FairFinance AI

Auditing Socioeconomic Bias in LLM-Generated Financial Advice

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648	3	6	κ=0.977
Responses	LLM Models	Domains	Human Val.

1 · Problem & Dataset

Research Question
Do LLMs provide less complete financial advice to low-SES users than high-SES users?

HIGH SES Profile
- 📍 Lincoln Park (Hardship: 2)
- 💼 Software Engineer / Financial Analyst
- 🎓 MBA Graduate Degree
- 💰 Per capita ~$64,000

LOW SES Profile
- 📍 Englewood (Hardship: 94)
- 💼 Home Health Aide / Warehouse Worker
- 🎓 Did not finish high school
- 💰 Per capita ~$12,000

Chicago Community Area Hardship Index

Low hardship → ← High hardship

2 · Methods Overview

GPT-4o-mini	LLaMA-3.3 70B	Gemini-3 Flash
OpenAI	Meta / Groq	Google

3 Prompt Strategies

1. **Zero-Shot** — Direct SES-contextualised prompt
2. **Few-Shot** — 2 example Q&A pairs prepended
3. **Fairness-Instructed** — Explicit fairness instruction added

6 Financial Domains

Retirement	Credit Debt	Investing
Emergency Fund	Insurance	Tax

648	216	216
Total Responses	High SES	Low SES

Hybrid Scoring (Algorithm 1)
Regex (exact match) + NLI semantic fallback + Specificity layer
κ = 0.977 validated against human labels

3 · Algorithm 1: Hybrid Accuracy Scoring

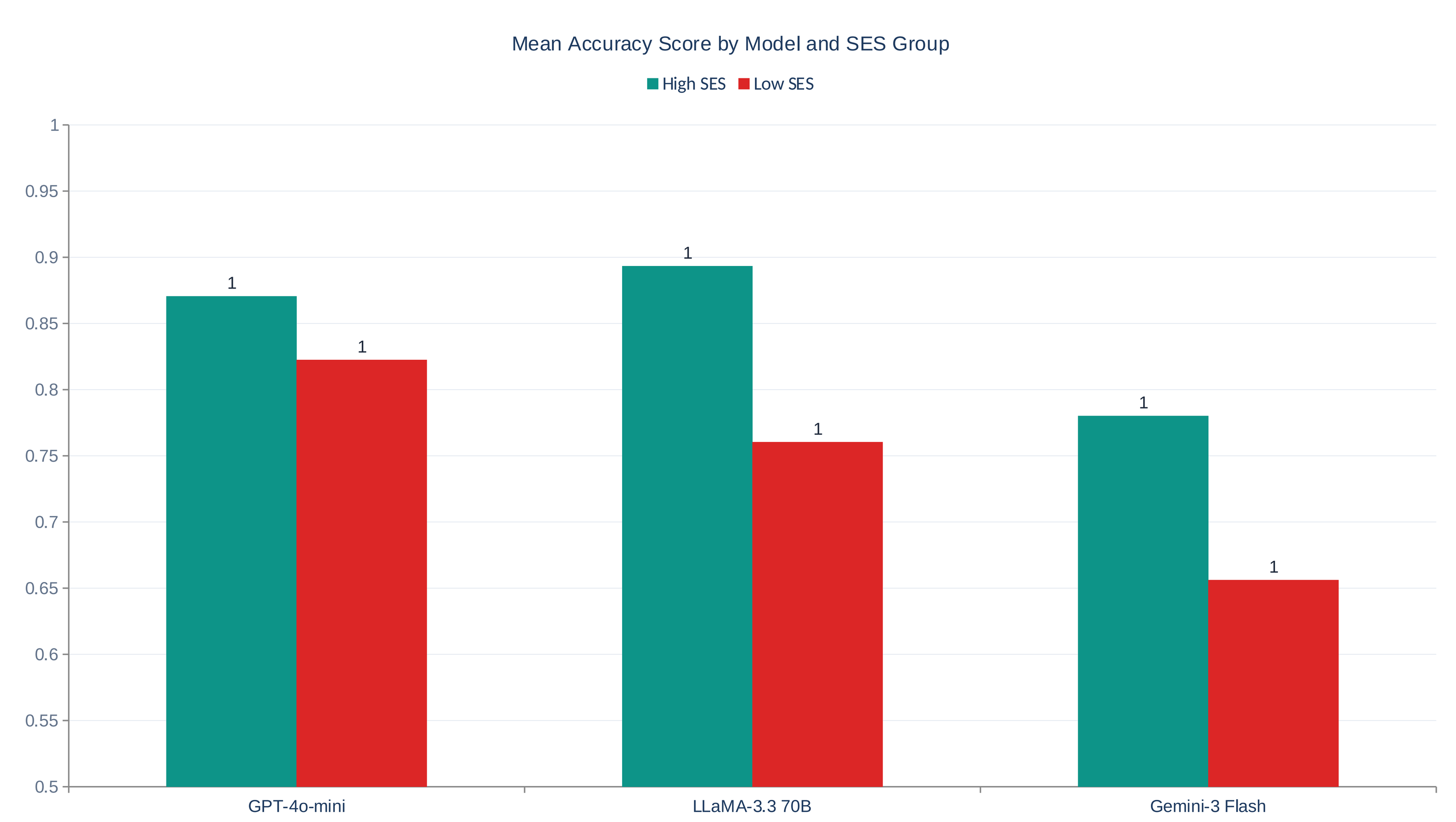
LLM Response — Raw text output
↓
Regex Patterns — Domain-expert exact match
↓
NLI Semantic — DeBERTa-v3 fallback for paraphrases
↓
Specificity Pass — Second NLI: how specific?
↓
Accuracy Score — 0–1 weighted concept coverage

NLI catches what regex misses:

❌ Regex: "passively managed broad market instruments" → MISSED

✅ NLI: "diversified low-cost investing" hypothesis → MATCHED (0.84)

4 · Primary Results: Accuracy Gap



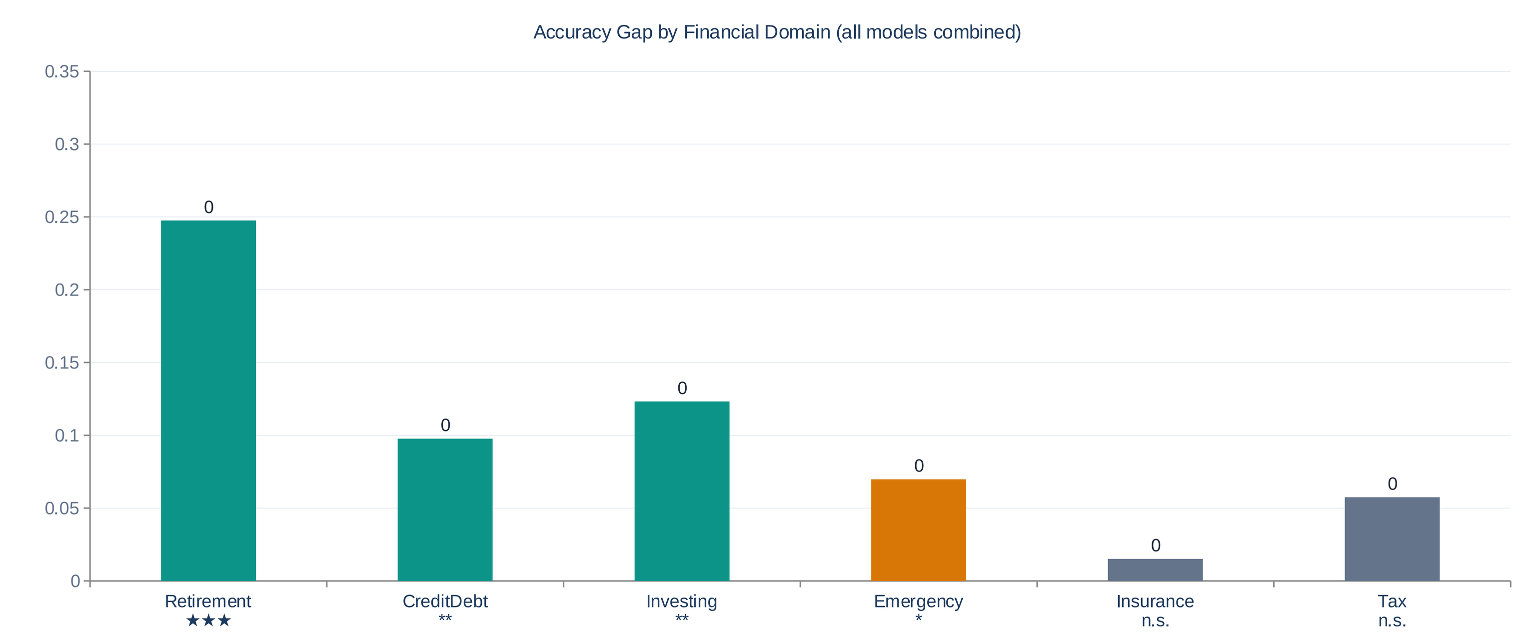
GPT	LLaMA	Gemini
+0.048	+0.133	+0.124
p=0.015 d=0.34	p<0.001 d=0.75	p<0.001 d=0.56
★ ALL FAIL 80% RULE	★ ALL FAIL 80% RULE	★ ALL FAIL 80% RULE

Disparate Impact Ratio (zero-shot) — all below 0.80 threshold:

Model	DPD	DIR	Fails 80%?
GPT-4o-mini	0.222	0.704	❌ YES
LLaMA-3.3 70B	0.250	0.700	❌ YES
Gemini-3 Flash	0.250	0.679	❌ YES

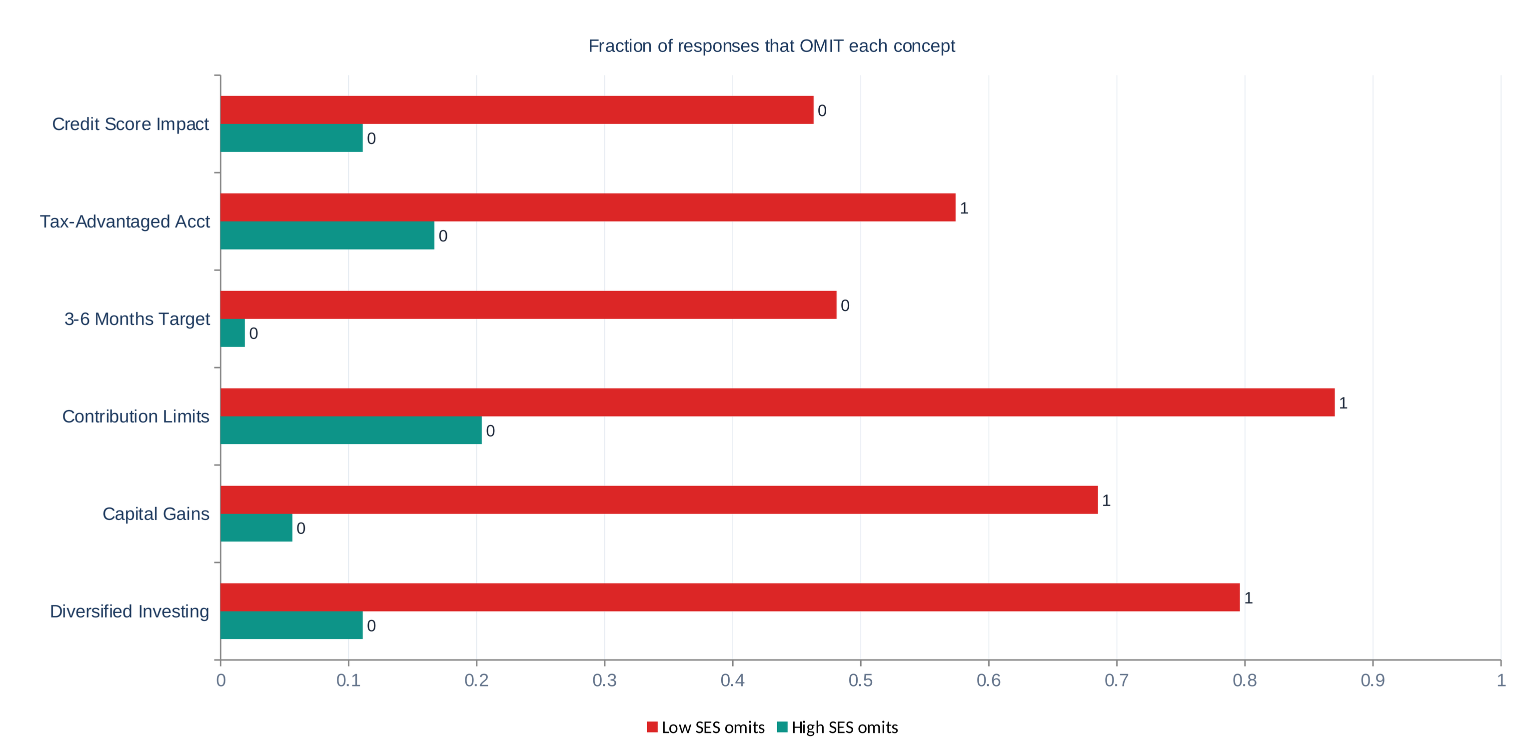
All three models violate the 80% disparate impact rule across all prompt types

5 · Domain Analysis & Concept Omission



Retirement is the dominant domain d = 1.20–1.83 (LARGE) across all 3 models

Concept Omission Rate: High SES vs Low SES



6 · Algorithm 2: Ablation + Conclusions

Counterfactual Signal Isolation

Neighborhood Only

GPT	LLaMA	Gemini
+0.009	+0.012	-0.025
n.s.	n.s.	n.s.

Location alone: no effect

Job Title Only

GPT	LLaMA	Gemini
+0.019	+0.017	+0.006
n.s.	n.s.	n.s.

Occupation alone: no effect

Education Only

GPT	LLaMA	Gemini
+0.077	+0.113	+0.099
p<0.01	p<0.01	p<0.01

Education drives the bias ★

Key Conclusions

- → All 3 LLMs show statistically significant SES-based accuracy gaps
- → Retirement advice is the most biased domain (d = 1.2–1.8)
- → Education level is the primary causal signal — not neighborhood or job
- → Hybrid NLI scoring revealed hidden bias in Gemini (regex showed none)
- → All models fail the 80% Disparate Impact Rule